Supplementary Material: Hallucination in Large Language Models: A Configuration-Level Study on TriviaQA

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This document provides intermediate results and implementation details that support the main report. It is organised into five sections corresponding to the key methodological decisions made in this study.

1 Automatic vs Manual Label Precision Analysis

Table 1 reports the full precision analysis from the 208 manually annotated cases. For each automatic label category, precision is defined as the fraction of cases where the automatic label matches the human-assigned label. The manual label breakdown column shows where disagreements went — revealing the specific failure modes of the automatic rule.

Table 1: Precision of automatic hallucination labels based on 208 manually annotated cases. Precision = proportion of auto-labelled cases where the human annotation agreed. Manual breakdown shows where disagreements were reassigned.

Auto Label	**n**	**Agree**	**Precision**	**Manual breakdown**
Hallucinated	104	90	0.865	90 Hallucinated, 9 Correct, 5 Abstained
Correct	56	54	0.964	54 Correct, 2 Hallucinated
Uncertain	40	3	0.075	19 Correct, 18 Hallucinated, 3 Uncertain
Abstained	4	2	0.500	2 Abstained, 1 Correct, 1 Hallucinated
Confident-Wrong	4	0	0.000	4 Hallucinated
Overall	**208**	**149**	**0.716**	—

Key findings. The Uncertain category is almost entirely unreliable: only 3 of 40 auto-Uncertain cases were confirmed as Uncertain by manual inspection. The remaining 37 split almost evenly between Correct (19 cases, 47.5%) and Hallucinated (18 cases, 45.0%), meaning the automatic rule provides no discriminative signal for this label. This finding directly motivates the correction formula used in the main report:

$$CorrectedHall. = (Auto - Hall \times 0.865) + (Auto - Uncertain \times 0.925)$$

where 0.925 = 1 $-$ 0.075 captures hallucinations hiding in the Uncertain bucket.

Failure modes. Manual inspection of the 59 disagreement cases revealed four systematic failure patterns: (i) synonym mismatches where the model answered correctly in different words (*e.g.* "permitted" vs "permissible", "housefly" vs "fly"); (ii) specificity mismatches where the model gave a more precise answer than the gold label required (*e.g.* "Puff the Magic Dragon" vs gold "Puff"); (iii) abstention-disguised-as-failure where the model said "not mentioned in the passage" which the automatic rule counted as Hallucinated; and (iv) alias coverage gaps where the gold alias list simply did not include the model's correct surface form.

2 Per-Configuration Manual Annotation Rates

Table 2 reports the manual label distribution for each of the eight model/configuration combinations, based on the stratified 208-case annotation sample. These rates are more reliable than the automatic hallucination rates for characterising true model behaviour, because they are not confounded by the alias coverage and synonym issues described above.

Table 2: Manual annotation rates per model and configuration. n_sampled is the number of cases drawn from that cell. All rates are percentages of n_sampled.

Model	Config	n	Correct (%)	Hall. (%)	Abstained (%)	Uncertain (%)
3B	Config 1 (closed-book)	25	44.0	52.0	0.0	4.0
3B	Config 2 (abstention)	27	44.4	48.1	7.4	0.0
3B	Config 3 (RAG)	25	28.0	72.0	0.0	0.0
3B	Config 4 (confidence)	27	40.7	55.6	3.7	0.0
8B	Config 1 (closed-book)	25	32.0	68.0	0.0	0.0
8B	Config 2 (abstention)	27	55.6	44.4	0.0	0.0
8B	Config 3 (RAG)	25	40.0	44.0	12.0	4.0
8B	Config 4 (confidence)	27	33.3	59.3	3.7	3.7

Notable observations. The 3B Config 3 (RAG) manual hallucination rate of 72.0% is the highest of any configuration, substantially higher than the 3B closed-book rate of 52.0%. This suggests BM25 retrieval actively confuses the smaller model by introducing irrelevant context it cannot distinguish from supporting evidence. The 8B Config 2 (abstention) achieves the lowest manual hallucination rate at 44.4%, confirming that the abstention prompt is the most effective single intervention for the larger model. The 8B Config 3 (RAG) rate of 44.0% is comparable, though with 12.0% of cases marked Abstained — the model frequently responded "not mentioned in the passage" when the retrieved passage was irrelevant.

3 Configuration 4 (8B): Format Collapse Breakdown

The 8B model under Configuration 4 exhibited a severe format compliance failure that inflated its reported hallucination rate. Table 3 provides a precise breakdown of output types for this configuration.

Table 3: Output type breakdown for 8B Configuration 4 (17,944 total outputs). Unparseable outputs are counted as Hallucinated by the automatic rule because the extractor returns an empty string, which scores EM=0 and F1<0.3 by construction.

Output type	Count	Percentage	Counted as
No parseable confidence rating	4,037	22.5%	Hallucinated (auto rule)
Confidence parsed, answer empty	6,110	34.0%	Hallucinated (auto rule)
Total unparseable	**10,147**	**56.5%**	—
Parseable (confidence + answer)	7,797	43.5%	Scored normally
— of which: EM=1 (Correct)	2,106	27.0% of parseable	Correct
— of which: Hallucinated	3,509	45.0% of parseable	Hallucinated
— of which: Uncertain	2,182	28.0% of parseable	Uncertain
Total	**17,944**	**100%**	—

Interpretation. Among the 43.5% of outputs that are parseable, the EM score rises to 0.270 and the hallucination rate falls to 45.0% — figures far more comparable to the other configurations. The 56.5% unparseable rate is not a hallucination phenomenon but a format compliance failure: the model generates content but loses the structured "Confidence: [H/M/L] Answer: [...]" format. Typical failure patterns observed in manual inspection include: generating underscore templates ("_____ Answer: _____"), generating numbered lists of multiple confidence-answer pairs, and repeating the question instead of answering it. The same prompt produces 99.7% parseable output from the 3B model, confirming this is a scale-dependent instruction-following failure rather than a knowledge failure.

4 BM25 Retrieval Corpus Construction

This section provides full implementation details for the BM25 retrieval pipeline used in Configuration 3, which are summarised in the main report but not given in full detail.

4.1 Corpus construction

The TriviaQA RC dataset includes a `search_results` field for each question, containing a list of web search result passages stored under the `search_context` key. For corpus construction, we extracted the first valid (non-empty) passage from each question's `search_context` list. If a question had no valid passage, the question text itself was used as a fallback (extremely rare in practice). This produced a fixed-size corpus of exactly 17,944 passages — one per question.

4.2 Index construction and retrieval

The corpus was tokenised at whitespace level (no stemming, no stop-word removal, no lemmatisation) and indexed using `BM25Okapi` from the `rank-bm25` Python library. For each question at inference time, the question text was tokenised identically and scored against all 17,944 corpus passages using the BM25 scoring function. The top-1 scoring passage was selected and truncated to 300 words before insertion into the prompt.

4.3 Zero fallback guarantee

All 17,944 retrievals were pre-computed before the inference loop, with a verification check confirming zero empty passages. This contrasts with the original implementation which searched only within each question's own passage list and fell back to closed-book when that list was empty (46.9% of questions). The new implementation eliminates this fallback entirely, making Configuration 3 a pure RAG condition across all questions.

4.4 Retrieval limitations

BM25 is a lexical retriever and cannot match passages to questions that use different vocabulary to express the same concept. On TriviaQA, many questions use formal phrasing while passages use informal phrasing or vice versa. In addition, the corpus is built from TriviaQA's own search result passages, which were collected using the gold answer as a search query — creating a potential circularity where the retrieved passage may already contain the answer regardless of retrieval quality. A stronger study would retrieve from an independent Wikipedia dump using the question text only.

5 Full Experimental Parameters

Table 4 provides a complete reference for all experimental parameters used in this study. These are stated individually in the main report but collected here for convenience.

Table 4: Complete experimental parameters.

Parameter	**Value**
Models	
Small model	Llama-3.2-3B-Instruct
Large model	Llama-3.1-8B-Instruct
Model precision	float16
Device	NVIDIA A100-SXM4-80GB (80GB VRAM)
Inference	
Batch size	16
Max new tokens	50
Decoding strategy	Greedy (do_sample=False)
Padding side	Left
Random seed	42
Dataset	
Dataset	TriviaQA RC validation split
Total questions	17,944
Mean answer length	2.1 tokens
Mean alias count	13.6 per question (range 1–173)
Evaluation	
EM normalisation	Lowercase, remove punctuation, remove articles
F1 normalisation	Same as EM
F1 strategy	Maximum F1 over all aliases
Retrieval (Config 3)	
Retriever	BM25Okapi (rank-bm25 library)
Corpus size	17,944 passages
Top-k	1
Passage truncation	300 words
Tokenisation	Whitespace split, lowercase
Closed-book fallback	None (zero fallback)
Hallucination labelling	
Correct	EM = 1
Hallucinated	EM = 0 AND F1 $<$ 0.3 AND NOT abstained
Uncertain	EM = 0 AND F1 $\geq$ 0.3 AND NOT abstained
Abstained	Output matches abstention pattern
Confident-Wrong	Config 4 only: High confidence AND EM = 0
Correction factors (from 208 manual annotations)	
Hallucinated precision	0.865
Uncertain precision	0.075
Correct precision	0.964
Abstained precision	0.500
Manual annotation	
Total cases	208
Sampling strategy	Stratified by model, config, and auto label
Per-cell target	13 Hallucinated, 7 Correct, 5 Uncertain + 2 Abstained (Config 2), + 2 Confident-Wrong (Config 4)
Random seed	42